
REGION RESONANCES

Methodology and Interpretation

Companion Document • 59 wine regions • semantic similarity

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1. MOTIVATION

Wine regions are conventionally indexed by geography, grape, and appellation. These categories are useful for navigating the world’s wine but they obscure a different dimension along which regions can be compared: the cultural and sensory *character* of the place itself — the temperament that shapes how its wines come to be made, marketed, and remembered. *The Soul of Wine* project encodes this character explicitly. Each of fifty-nine regions is assigned a one-word metaphor (Burgundy is *Devotion*, Bordeaux is *Business*, Santorini is *Survival*, Mosel is *Poetry*) and a roughly three-hundred-word identity narrative that develops the metaphor in context.

Region Resonances is a tool for navigating this character space by query. The user submits a word, phrase, or fragment of prose — *resilience, old soul, the feeling of late summer* — and the tool returns the regions whose written character is most similar to the query in a precise, computable sense. The aim is neither classification nor retrieval in the conventional database sense; it is closer to what an attentive reader of the project’s prose might suggest if asked which regions a given phrase reminded them of.

A useful analogy is a tuning fork: strike a note, and the objects in the room that are tuned to nearby frequencies will resonate. Region Resonances does the same with text. Type a phrase, and the regions whose narratives are tuned to nearby semantic frequencies will respond.

2. WHAT THE MATCHER READS

The tool operates on two pieces of canonical input per region, and only on those two pieces. First, the one-word metaphor. Second, the full identity narrative for that region (typically 280–320 words). Together they form a single passage of text that captures, in compressed form, the project’s claim about what the region *is*.

The matcher does not read grape varieties, soil types, vintage notes, scores, or any other conventional descriptor. It does not consider what a region’s wines are like to drink. It considers only what the region itself is like in character, as expressed in the project’s own prose. This is a

deliberate scoping decision. Region Resonances is a tool of identity, not of palate; it complements the existing *Region Affinities* tool rather than overlapping with it.

Both data sources — the metaphor list and the narrative text — are locked. They are not regenerated, modified, or re-tuned at runtime.

3. THE MATCHING PROCEDURE

The procedure has three stages: representation, comparison, and ranking.

3.1 Representation

The matcher does not work directly with text. Instead, every passage of text — both each region’s narrative and the user’s query — is first converted into a numerical representation called a *semantic embedding*. An embedding is a list of approximately fifteen hundred numbers, computed by a large neural language model that has been trained to assign similar number-lists to passages with similar meaning.

The crucial property of a semantic embedding is that it abstracts away from surface vocabulary. Two passages can share no common words and still receive nearly identical embeddings if their meaning is close. Conversely, two passages can share many common words — think of two sentences differing only by a negation — and receive quite different embeddings. The embedding captures meaning at a level above the words themselves.

For Region Resonances, every region’s metaphor-and-narrative passage was embedded once, in advance, and the resulting fifty-nine number-lists were saved to a small file shipped with the tool. This pre-computation step has no runtime cost.

3.2 Comparison

When the user submits a query, the matcher embeds the query using the same model that produced the region embeddings, yielding another list of approximately fifteen hundred numbers. The query and each region are now represented in the same space, and the question of *how similar are they?* becomes a geometric one: how aligned are the two number-lists?

The standard answer to this question is called *cosine similarity*. Two embeddings are treated as arrows pointing from the origin into a high-dimensional space; the cosine similarity between them is, geometrically, the cosine of the angle between the arrows. It ranges from -1 (opposite directions) through 0 (perpendicular, unrelated) to $+1$ (identical direction). For this corpus, the matcher computes the cosine similarity between the query and each of the fifty-nine region embeddings.

3.3 Ranking

The fifty-nine similarity values are sorted from highest to lowest, and the top five are presented to the user as the resonance result. The remaining fifty-four are accessible through a *Show all 59 ranked* disclosure for users who want to see the full neighbourhood.

The result is therefore a ranked list, not a verdict. The top match is the region whose narrative is most semantically similar to the query, but the second, third, and fourth matches are often equally informative — and sometimes more interesting, because the top result is occasionally the one a thoughtful reader would have predicted.

4. INTERPRETING THE RESONANCE SCORE

Each match in the result list carries a numerical score between 0 and 100. The score is a presentation transformation of the underlying cosine similarity, scaled so that the typical range of useful matches occupies most of the visible scale. A score of 84 indicates that the region’s narrative sits very close to the query in semantic space; a score of 51 indicates a more distant but still detectable resonance.

Two interpretive cautions apply. First, the score is *comparative, not absolute*. A score of 70 does not mean the region is “70% the query” in any meaningful sense. It means the underlying embeddings are aligned to a degree that places the region in the upper portion of the corpus relative to that query. The score is most informative when read in context with the other scores in the same result list.

Second, perfect or near-perfect scores are essentially impossible unless the query is itself drawn from a region’s narrative. The matcher is not designed to discover identity; it is designed to discover resonance. A top match with a score in the seventies typically reflects strong, genuine semantic kinship.

5. STRENGTHS AND LIMITATIONS

The matcher works best with queries that carry semantic content of the kind the corpus describes: temperament, posture, mood, cultural attitude. Queries such as *aspiration*, *stubborn quiet*, or *the feeling of late summer* produce satisfying results because they engage the same descriptive register the narratives use. Such queries are also where the abstract-meaning capability of the embedding model is most useful; the matcher routinely returns regions whose narratives share no surface vocabulary with the query but whose character clearly resonates with it.

The matcher is poorly suited to queries that ask for facts the corpus does not contain. The narratives describe regional character; they do not describe grape varieties, climate data, or production volumes. A query such as *What grapes does Mosel grow?* will still produce a ranked list, because the matcher always produces a ranked list, but the ranking will be uninformative. This is a scoping limitation, not a failure of the underlying technology.

A subtler limitation concerns variation in narrative length and thematic range. Regions whose narratives interleave several themes (Tokaj, for instance, weaves melancholy, decline, and inherited beauty) tend to appear in the top results for a wider variety of queries than regions whose narratives are tightly focused. We have not corrected for this. The bias is mild and arguably desirable: a region of greater thematic range *should* resonate with more inputs.

Finally, the system is monolingual. The embedding model can process input in many languages, but the corpus it is matched against is English. Queries in other languages will return ranked results, but the rankings will be biased and should not be relied upon.

6. WORKED EXAMPLES

The following examples are drawn from the calibration test set used during development. They illustrate the kind of result the matcher produces.

The query *old soul* returns Burgundy (*Devotion*), Rioja (*Patience*), Tokaj (*Melancholy*), and Sardinia (*Stubbornness*) in the top four positions. None of those metaphors contain the word *old* or the word *soul*; the match is operating at the level of meaning. All four regions are characterised in the corpus by inherited tradition, slow time, and a posture of waiting rather than acting.

The query *resilience* returns Santorini (*Survival*), Barossa Valley (*Fortitude*), Douro (*Endurance*), and Finger Lakes (*Conviction*). Again, no surface-word overlap; the embedding places the query close to the cluster of regions whose narratives describe sustained effort against adverse conditions.

The query *the feeling of late summer* returns Tokaj (*Melancholy*), Provence (*Pleasure*), and Galicia (*Longing*). This is a more impressionistic query and the matches reflect that — the response includes a melancholy region, a sensory-pleasure region, and a region defined by emotional distance, all of which are plausible interpretations of the seasonal mood the query evokes.

These examples are not retrievals in the database sense. They are interpretations of the query against a fixed body of descriptive prose, mediated by a model that has internalised the structure of meaning in English. The matches are repeatable (the same query always returns the same list), but they reflect the meaning of the corpus, not facts about wine.

7. PRIVACY

Queries are transmitted to the embedding service for the purpose of computing the semantic embedding. They are not stored, logged, or aggregated. There are no user accounts and no session-spanning history. The same query submitted twice produces the same result, but this is a property of the model and the corpus, not of any record kept of the user’s previous activity.

For the mathematical treatment of the procedure, see the accompanying Technical Appendix.